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Learning as a Generative Process

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A cognitive model of human learning with understanding is introduced. Empirical research supporting the model, which is called the generative model, is summarized. The model is used to suggest a way to integrate some of the research in cognitive development, human learning, human abilities, information processing, and aptitude-treatment interactions around the notion of transfer of experience and abilities.

Since the turn of the century, educational psychology has led in producing fundamental changes and paradigmatic shifts in the study of human learning. E. L. Thorndike greatly affected our conception of human learning when he successfully extended his laboratory findings into empirical research in education. As a result, he devastated faculty psychology, and he revolutionized our conceptions of transfer, learning, teaching, and instruction.

B. F. Skinner, Robert Gagné, Richard Atkinson, Robert Glaser, Patrick Suppes, and other researchers in learning in large part produced their outstanding contributions by developing and extending their ideas into education, including training, teaching, and instruction. As is the case with creativity in other scientific fields, a paradigm becomes accepted and changes the field when it stands tests of utility within and outside the laboratory. For people who study learning, one of the best places to demonstrate the utility of their ideas is in education.

Within the last decade or so, educational psychology has again been very much involved in producing a paradigmatic shift in the study of human learning. The current welcomed shift is toward cognitivism, toward reinstating the learner, and his cognitive states and information-processing strategies, as a primary determiner of learning with understanding and long-term memory.

The shift is welcome for two reasons. It promises fundamental advances in understanding human learning and in integrating findings from developmental psychology, human abilities, and human learning. Secondly, it is time to accept some of the points of reinforcement theory and to move on to other important issues. These issues include the cognitive processes of learners, organization, proactive variables, the relationships between aptitudes and instruction, and the relations between stored knowledge and the generation learning.

The reasons for the latest paradigmatic shift in the study of learning are complex. At a deep level they probably involve a reaffirmation of the dignity and freedom of individual human beings in a mass technological society; of concern over the long-range effects upon developing learners; of holding the environment, rather than the learners, responsible for their behavior; of performance contracting; of accountability only for teachers; of behavioral objectives; and of related techniques, which also attribute causes of behavior to the environment.

But more specifically, in the study of learning the paradigmatic shift is coming about because of the advantages obtained by replacing nominal stimuli with functional ones; i.e., by introducing into the paradigms for studying learning the perception and interpretation of the learner processing the information and actively constructing meaning from it. Consider the following recent empirical results: on organization (e.g., Ausubel, 1960; Bower, 1972; Mandler, 1967; Neisser, 1967); information processing, encoding, and

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memory (Atkinson, 1973; Tulving & Thomson, 1973); imagery (Paivio, 1971); the delay-retention effect (Sassenrath & Yonge, 1969); cognitive reinterpretations of attribution in motivation (Weiner, Heckenhausen, Meyer, & Cork, 1972); the lateralization of brain functions (Sperry, 1968); proactive processes (Underwood, 1957); personality (Mischel, 1973); and even behavior modification (Bandura, 1969, p. 127). In these studies, it is the learner's interpretation of and processing of the stimuli, not their nominal characteristics, which is primary.

Educational psychology should be and will again be the crucible in which new directions in human learning are tested and developed. The meaning of a new paradigm of learning becomes real and concrete when it is studied in terms of human beings learning organized information. I think we should take the lead in revising and advancing our fundamental conceptions of how types of learning occur and are facilitated.

From my research on transfer, I will now try to show how one cognitive model, which I call the generative model, indicates some new directions for studying several phenomena in human learning. Last I will suggest a way to integrate some of the research in human development, human learning, abilities, and aptitude-treatment interactions around the notion of transfer of prior experience.

THE GENERATIVE MODEL OF LEARNING

To introduce the generative model, let me begin by stating its fundamental premise, which is that people tend to generate perceptions and meanings that are consistent with their prior learning. For example, an operant conditioner who for years has been observing rats in a Skinner box tends to perceive the process of children learning in school as the process of bringing their behavior increasingly under stimulus control through carefully sequenced, successively approximated, discriminatively reinforced contingencies. A cognitivist, who has observed too many children learning in school, may, upon witnessing a hungry rat learning to press the bar to get food, perceive the lowly hooded rat as an independent, autonomous learner, using his previously acquired information-processing strategies to construct adaptive, even creative responses to solve the personally important problem of getting food when one is hungry and alone in a foreign, technologically complex environment.

Neither zeitgeist is true or false, only more or less productive for a given problem. There is, after all, a subtle but most significant difference between "hours of food deprivation" and the mental state called "hunger." Hunger can go down with increasing hours of deprivation, if the experimenter pleads with the subject to persist a little longer. Or hunger can instantly increase with a cleverly prepared, illustrated description of delicious food. The relevant issue is whether the nominal stimulus or the functional stimulus has greater utility for research in educational psychology.

In contrast to earlier concepts of meaningfulness or semantic-processing models of encoding and learning, the generative model predicts that learning is a function of the abstract and distinctive, concrete associations which the learner generates between his prior experience, as it is stored in long-term memory, and the stimuli. Learning with understanding, which is defined by long-term memory plus transfer to conceptually related problems, is a process of generating semantic and distinctive idiosyncratic associations between stimuli and stored information.

Although the model is related to other cognitive models of learning, such as Tulving's encoding specificity model and Paivio's dual processing model, it is unlike any of them in its emphasis upon both concrete, prior, distinctive experience and abstract verbal abilities transferring to and being used to construct specific associations to stimuli.

From the above model, my students and I have conducted a number of studies. I will briefly describe three of them, and then move to the larger issues.

The data of the two experiments of the first study are reported in Figure 1. Three hundred and sixty-six fifth- and sixth-grade public elementary school children, 186 in the below-average and 180 subjects in the above-average reading level experiments, respectively, were randomly assigned to the nine conditions given on the abscissa of Figure 1. In the three control groups, indicated by a capital C, plus a subscript, the children read passages from commercially published, widely used reading materials or from control materials. The groups given one or two word organizers at the top of each paragraph in the text are indicated by a capital O with the numeral 1 or 2, respectively, as a subscript. The three groups asked to generate and write a sentence summarizing the meaning of each paragraph are indicated with an uppercase G. The groups given both a one- or two-word organizer and directions to generate a sentence to describe the meaning of each paragraph are indicated by capital GO.

I predicted the generative groups to perform the best, the organizer groups next, and the control groups last. As Figure 1 indicates, retention and comprehension increased sizably and statistically significantly as predicted, and, without exception, in the order predicted. One group, GO_2, doubled the scores of the control group that read the same stories without generative instructions or organizers.

The next figure, Figure 2, reports an experiment which John Carter and I conducted with college students asked to free recall hierarchically arrayed terms that were (a) unrelated words, (b) randomly arranged conceptually related words, and (c) properly arranged conceptually related words. The learners were given two different directions, either (1) to generate hierarchies that "made sense," or (2) to copy the words in the hierarchies they were given. Unlike semantic-processing models, the generative model predicts not only a facilitation of free recall with the generative instructions, but the greatest facilitation in the group that could generate both distinctive associations and abstract ones, i.e., the random group. We predicted that the unrelated group would generate

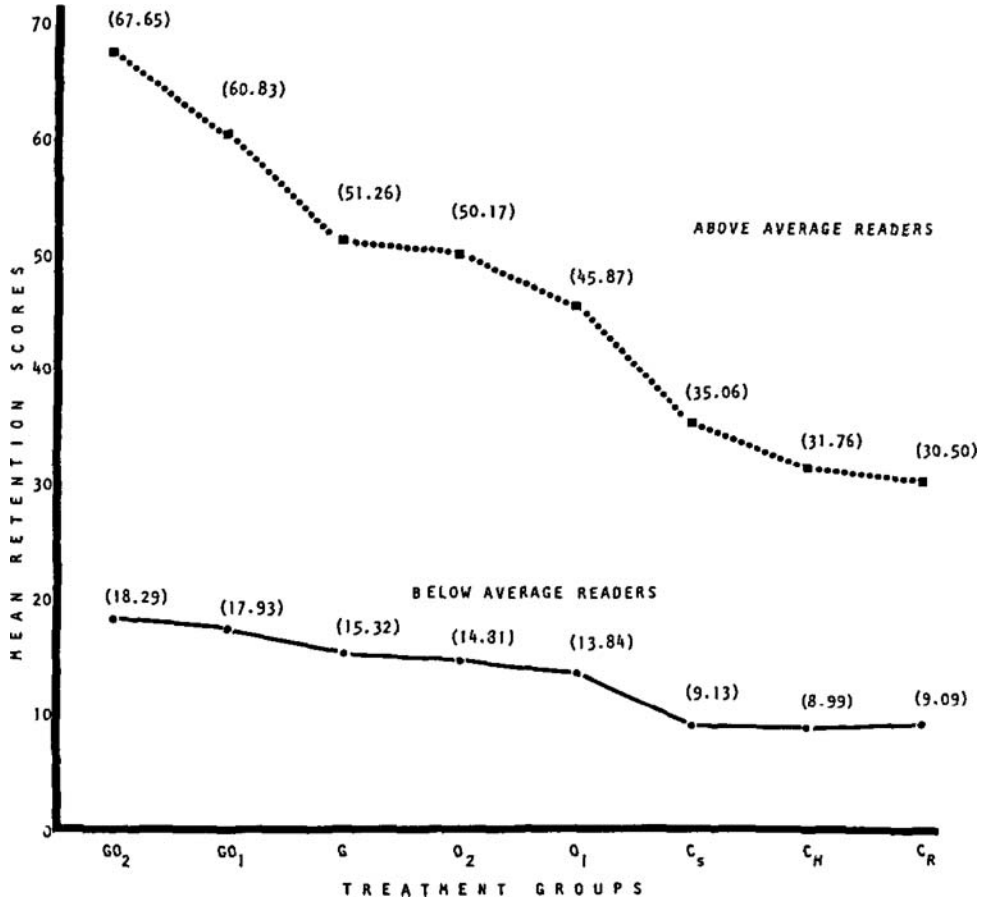


FIGURE 1 Mean retention scores for above-average and below-average readers. (Adapted from Doctorow, Wittrock, & Marks, 1978.)

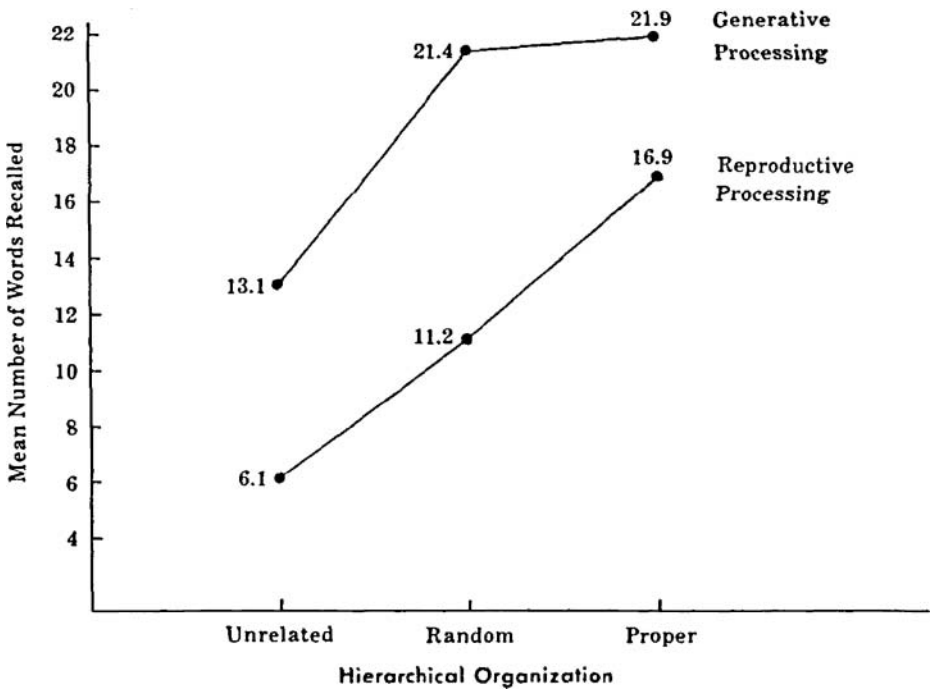


FIGURE 2 Results from Wittrock and Carter (1975) study.

TABLE 1
Planned Comparisons Among the Mean Errors Transfer Test I: Experiment II

	Treatment Group							
	D_{1R} *(Shape)*	D_{1R} *(Brightness)*	D_{2R} *(Shape)* *(Brightness)*	D_{1Ir} *(Border)*	D_{1Ir} *(Number)*	D_{2Ir} *(Border)* *(Number)*	*Control*	*T*
Mean Errors	9.33	.68	2.28	27.30	12.29	18.52	9.14	
N	21	19	18	20	21	21	22	
1. One Problem vs. Two Problems	−.25	−.25	.5	−.25	−.25	.5	0	
2. Shape vs. Brightness	1	−1	0	0	0	0	0	3.23**
3. Two Relevant vs. Two Irrelevant	0	0	1	0	0	−1	0	3.70**
4. One Relevant vs. Two Relevant	−.5	−.5	1	0	0	0	0	

** $p < .01$.

only distinctive, concrete associations from their long-term memory because there were no abstract, conceptually related words in the higher levels of the hierarchy. The proper group was predicted to generate primarily abstract, semantic associations before determining that the given hierarchy was a useful one. The data supported the predictions ($p < .001$). Even asking people to try to generate a hierarchy among words chosen at random from a dictionary doubled their free recall.

One example is needed to show how the generative model relates development to learning. Table 1 presents the data from 140 fifth-grade public school children who learned to solve simple concept identification problems involving shape, brightness, border, and number. Shape and brightness were the relevant dimensions in the bidimensional concept problems. That is, the correct answer always involved attending to both brightness and shape, such as choosing the black circle.

From the generative model, we predicted that the training should focus on rewarding the children for choosing the stimuli by their nonpreferred but still relevant dimension, in this case brightness. As is typical of children of their age, with very few exceptions these 11-year-old children were found in the pretest to prefer shape over brightness.

The data closely support the model. Training on the nondominant but relevant dimension is all that is needed to obtain almost perfect transfer to the two-dimensional problems, where the first response must be by chance, with a probability of .5 of being correct. Again, it is the relation between the learner's background and the stimuli which was crucially important in the design of the treatment. In this case, training people to select only one dimension produced slightly better transfer than training them by having them practice the actual behavior required on the transfer test.

TRANSFER DESIGNS AND REDIRECTING RESEARCH IN LEARNING

From the above data, I believe that the generation from one's memory of abstract and concrete associations is a cognitive process involved in learning with understanding, comprehension, and long-term recall. The generative model presented above has implications for reconceptualizing some research in instruction and for relating research in human abilities, development, and learning to one another.

The theme that relates all of these areas of research to one another is the notion that human learning with understanding involves the process of generating and transferring meaning for stimuli and events from one's background, attitudes, abilities, and experiences. As one example of the implications which follow from such a conceptualization, Larry Goulet and I have previously presented (Goulet & Wittrock, 1971) the figure which is reproduced here as Figure 3. The figure is Goulet's work (Goulet, 1970). This figure indicates several comparisons which would help to untangle previous experience at solving learning set problems from age differences in intellectual development. That is, to combine development research with research on learning, younger and older students are seen to vary not only in age but also in their prior experience with learning-to-learn tasks relevant to the

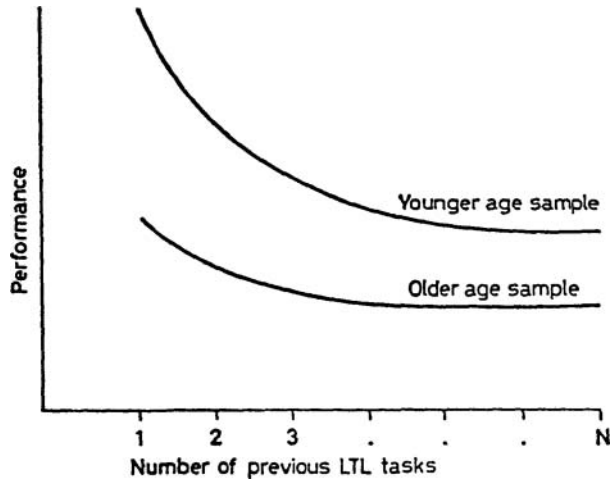


FIGURE 3 Hypothetical learning-to-learn (LTL) transfer functions for younger and older children. (Reproduced from Goulet, *Human Development*, 1970, 13, p. 234, courtesy of S. Karger AG, Basel.)

specific task in the experiment. To closely relate research on learning and development, we need to understand the effects of prior learning set experience. By giving the younger children preliminary trials at solving learning set problems, we can better understand their developmental differences. After these preliminary trials, we are ready to determine the developmental difference between the groups with regard to the specific task. Using transfer designs in this way leads one to construct types of transfer, from general to more specific processes of constructing associations.

FOR THE FUTURE

The generative model is only one example of my main point about how educational psychology should be using new ways to study learning and new ways to conceptualize important problems that cut across differential and experimental psychology. A shift to cognitive psychology does not mean an end to science in educational psychology. Instead, the shift means that the learners, especially their prior experiences, backgrounds, abilities, and attitudes, are part of our science and are crucially important in the equations for predicting learning with understanding. Cognitive theory implies that learning can be predicted and understood in terms of what the learners bring to the learning situation, how they relate the stimuli to their memories, and what they generate from their previous experiences.

Cognitive psychology has a potential for bringing together two schools of psychology, differential and experimental. Within the zeitgeist of cognitive psychology, aptitude-treatment interaction (ATI) research is a most important way to study learning, because ATI research is an effective way to study how different people generate associations by using their abilities and previous experiences. However, it is not only the *differences* in the learners' backgrounds as they imply *differences* in instructional treatments which are important for designing instruction. At a more basic level, the relations between the learners' backgrounds and the instruction are important for designing treatments. Even if all learners were to have identical backgrounds, which is of course not conceivable, the relation between their experiences and the instruction would still be crucially important, as it was in the study whose data appear in Table 1, where nearly all of the children preferred shape over brightness.

Viewing learning as a constructive cognitive process may also indicate one reason why the recent research on imagery and the recent research on semantic processing of information has each produced impressive results on memory. Imagery and semantic processing each sample a different part of the process of generating abstract and distinctive associations from one's memory. If we can devise ways simultaneously to stimulate generative processes, we may be able to develop additional increments in comprehension and free recall.

PROBLEMS IN CONDUCTING COGNITIVE RESEARCH

In sum, one new direction for research in educational psychology is the use of transfer designs to integrate research in development, human cognitive learning, abilities, and attitudes. A serious methodological problem in conducting empirical research on cognitive learning is the development of appropriate, useful indexes of the cognitive processes. Semantic processing has already acquired a circular definition in which the process and its result, i.e., long-term recall, are defined in terms of one another. If the responses were recalled from long-term memory, they are said to have been semantically processed. How do we know they were semantically processed? Because the responses were recalled from long-term memory. With circular definitions like that one, the next paradigmatic shift away from cognitive psychology cannot be far behind. From the last 70 years we should have learned the indispensable nature of scientifically measuring hypothetical processes and intervening variables independently from the dependent variables they are hypothesized to influence.

These methodological problems must not discourage us from pursuing the new directions indicated in the study of cognitive learning and memory. Through the study of cognitive and affective processes, we may better understand how to relate information to a learner's background and aptitudes, how to organize and sequence the information, and how to stimulate its meaningful processing, comprehension, and long-term memory. We may be able someday to understand how learning occurs without a large number of reinforced repetitions.

SUMMARY

Transfer designs are one important way to study how the learner, his experience and his cognitive processes, in large part, determine learning with understanding and long-term memory. Transfer designs bring the two worlds of psychology together. They also put the active learner foremost in hypotheses about learning and memory, indicating that the effects of instruction are understandable only in terms of what the instruction causes the learner to construct. From my research, it seems that instruction which causes the learner to generate distinctive associations between stimuli and memory facilitates long-term recall and understanding.

But regardless of which designs or theoretical approaches we use, I hope educational psychologists will continue to lead in the production of fundamental advances in the understanding of human learning and its facilitation.

REFERENCES

Atkinson, R. C. (1973). Ingredients for a theory of instruction. In M. C. Wittrock (Ed.), *Changing schools: Alternatives from educational research* (pp. 65–82). Englewood Cliffs, NJ: Prentice-Hall.

Ausubel, D. P. (1960). The use of advance organizers in the learning and retention of meaningful verbal material. *Journal of Educational Psychology, 51,* 267–272.

Bandura, A. (1969). *Principles of behavior modification.* New York: Holt, Rinehart, & Winston.

Bower, G. H. (1972). A selective review of organizational factors in memory. In E. Tulving & W. Donaldson (Eds.), *Organization of memory* (pp. 93–137). New York: Academic Press.

Doctorow, M., Wittrock, M. C., & Marks, C. (1978). Generative processes in reading comprehension. *Journal of Educational Psychology, 70*(2), 109–118.

Goulet, L. R. (1970). Training, transfer, and the development of complex behavior. *Human Development, 13,* 212–240.

Goulet, L. R., & Wittrock, M. C. (1971, April). *Transfer of training models and developmental processes*. Paper presented at the biennial meeting of the Society for Research in Child Development, Minneapolis, MN.

Mandler, G. M. (1967). Organization and memory. In K. W. Spence & J. L. Spence (Eds.), *The psychology of learning and motivation, Volume I* (pp. 327–372). New York: Academic Press.

Mischel, W. (1973). Toward a cognitive social learning reconceptualization of personality. *Pyschological Review, 80,* 252–283.

Neisser, U. (1967). *Cognitive psychology.* New York: Appleton.

Paivio, A. (1971). *Imagery and verbal processes.* New York: Holt, Rinehart, & Winston.

Sassenrath, J. M., & Yonge, G. D. (1969). Effects of delayed information feedback and feedback cues in learning and retention. *Journal of Educational Psychology, 60,* 174–177.

Sperry, R. W. (1968). Hemisphere deconnection and unity in conscious awareness. *American Psychologist, 23,* 723–733.

Tulving, E., & Thomson, D. M. (1973). Encoding specificity and retrieval. *Psychological Review, 80,* 352–373.

Underwood, B. J. (1957). Interference and forgetting. *Psychological Review, 64,* 49–60.

Weiner, B., Heckenhausen, H., Meyer, W. U., & Cork, R. E. (1972). Causal ascriptions and achievement behavior: A conceptual analysis of effort and reanalysis of locus of control. *Journal of Personality and Social Psychology, 21,* 239–248.

Wittrock, M. C., & Carter, J. F. (1975). Generative processing of organized information. Unpublished manuscript.